

What the Real Ensemble Package Measures —and What It Does Not

Cut Fraction Is Not Polsby–Popper

G.3 — Compactness Distribution Position

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Abstract

The active real ensemble package measures unweighted tract-graph cut fraction, not Polsby–Popper, Reock, perimeter, or the geographic boundary-length objective used by the BISECT benchmark. Rhode Island, Iowa, and North Carolina show low cut-fraction percentiles, but Iowa and North Carolina lack the ESS needed for precise tail claims and the two implementations disagree materially. No polygon compactness distribution was generated. The prior 61st–72nd Polsby–Popper percentile tables and compactness–partisan correlations are therefore withdrawn as empirical findings.

Contents

1	Metric Identity	3
2	Observed Positions	3
3	Implementation Sensitivity	3

4	Withdrawn Claims	3
5	Conclusion	4

1. Metric Identity

The real package records:

$$f_{\text{cut}} = \frac{\text{tract adjacency edges crossing districts}}{\text{all tract adjacency edges}}.$$

This metric ignores polygon perimeter, boundary length, projection, coastline, and area. It is a graph-topology fragmentation measure. The benchmark itself uses shared-boundary weights, so even its optimization objective is not identical to the unweighted evaluation metric [Polsby and Popper, 1991, Reock, 1961].

2. Observed Positions

Rhode Island’s benchmark is at the 1.35th percentile in Rust and 1.81st in GerryChain. Iowa is at the 0.33rd and 0.88th percentiles, with ESS 257 and 498. North Carolina is at 0 and 0.2 percent, with ESS 109 and 87. Under the preregistered rule, no sub-1% claim may be published unless ESS is at least 1,000. Thus only Rhode Island supports a stable bounded statement: the benchmark lies near the low end of the sampled cut-fraction distribution.

3. Implementation Sensitivity

The Rust and GerryChain kernels use the same random-weight Kruskal tree method, initial plan, population tolerance, and chain length, yet their cut distributions differ. The KS effect is small in Rhode Island (0.038), moderate in Iowa (0.162), and large in North Carolina (0.471). Proposal details, pair reselection, random streams, and implementation behavior therefore matter to distributional claims.

4. Withdrawn Claims

The package contains no polygon compactness trace. It cannot support:

- the former 61st–72nd Polsby–Popper percentile range;
- the claim that no benchmark exceeds the 95th compactness percentile;

- a compactness–partisan correlation; or
- a legal “optimal compactness position.”

Those questions require archived geometries or per-plan polygon metrics under a versioned projection and metric implementation.

5. Conclusion

Metric identity is part of evidence identity. A low tract cut fraction is a real, reproducible result for this package, but calling it “Polsby–Popper compactness” would be a category error.

References

- Daniel D. Polsby and Robert D. Popper. The third criterion: Compactness as a procedural safeguard against partisan gerrymandering. *Yale Law & Policy Review*, 9:301–353, 1991.
- Ernest C. Reock. A note: Measuring compactness as a requirement of legislative apportionment. *Midwest Journal of Political Science*, 5(1):70–74, 1961.